

RULES OF THUMB

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Big Picture

No unjustified 3D

Power of the plane
Disparity of depth
Occlusion hides
information
Perspective distortion
dangers
Tilted text isn't legible
Justification and
Alternatives
Benefits of 3D
Empirical Evidence

No unjustified 2D

Resolution over immersion

Overview first,
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Responsiveness
is required

Function first,
form next

Eyes beat
memory

- This lecture contains **rules of thumb**: *advice* and *guidelines*.
- These rules of thumb synthesize the current state of knowledge into a more unified whole.
- In some cases, they use empirical studies



No unjustified 3D

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- Disparity of depth
- Occlusion hides information
- Perspective distortion dangers
- Tilted text isn't legible
- Justification and Alternatives
- Benefits of 3D
- Empirical Evidence



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Responsiveness is required

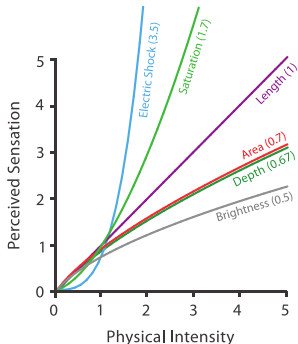
Function first, form next

Eyes beat memory

No Unjustified 3D

- Many people have the intuition that if two dimensions are good, three dimensions must be better
- There are **many difficulties** in visually encoding information with the third spatial dimension, depth.

Steven's Psychophysical Power Law: $S = I^N$





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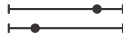
Function first,
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Eyes beat
memory

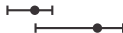
- The spatial position channels apply **only to** planar spatial position, **not** arbitrary 3D position.
- For the perceived importance of items ordered within the axes, reading conventions probably dominate.
 - most Western languages go from left to right and from top to bottom
 - but Arabic and Hebrew are read from right to left
 - some Asian languages are read vertically

➔ Magnitude Channels: Ordered Attributes

Position on common scale



Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)





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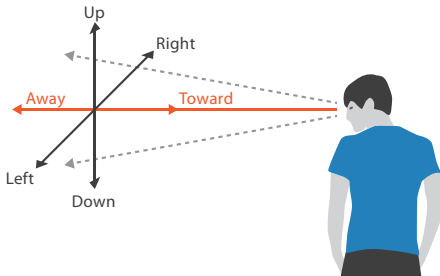
Responsiveness
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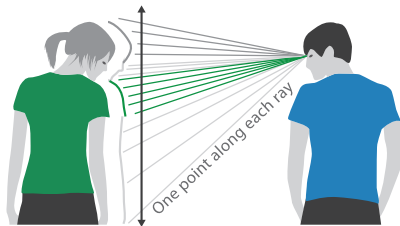
Eyes beat
memory

We do not really live in 3D, or even 2.5D: we **see** in 2.05D

- acquire more info on image plane quickly from eye movements
- acquire more info for depth slower, from head/body motion



Thousands of points up/down and left/right



We can only see the outside shell of the world



Occlusion hides information

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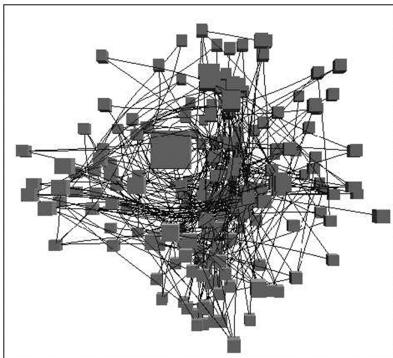
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- Occlusion
 - Some objects cannot be seen because they are hidden behind others.
 - The visible objects are interpreted as being closer than the occluded ones
- Interaction complexity





Perspective distortion dangers

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**Perspective distortion
dangers**

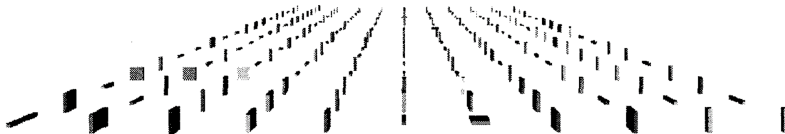
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- Perspective distortion loses information
 - interferes with all size channel encodings
 - power of the plane is lost!



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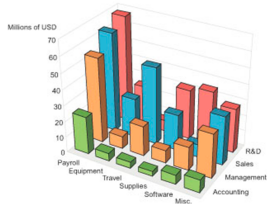
3D vs 2D bar charts

- 3D bars never a good idea!

Graph Design I.Q. Test

Question 7: Which graph makes it easier to determine R&D's travel expense?

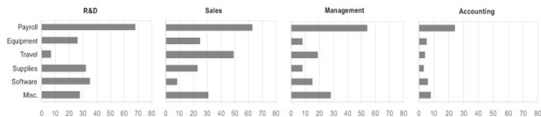
2006 Expenses by Department



3-D Bar Graph (left)

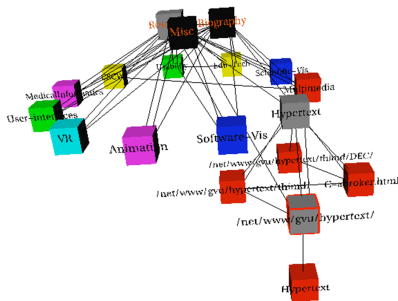
2-D Bar Graphs (below)

2006 Expenses by Department in Millions of USD



Text legibility

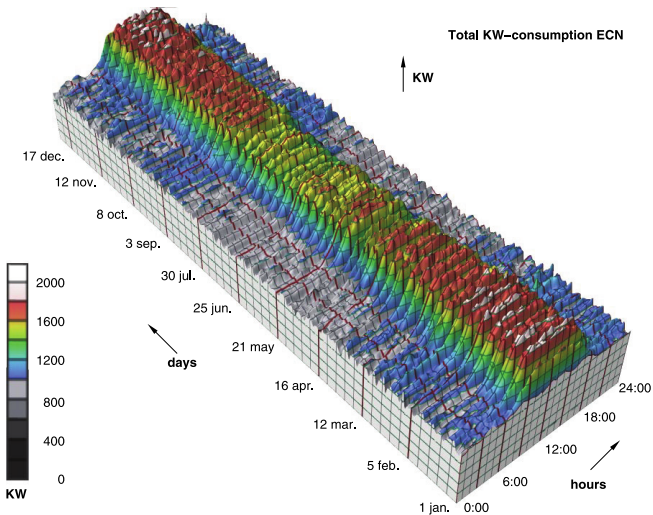
- Text fonts have been very carefully designed for maximum legibility when rendered on the grid of pixels that makes up a 2D display
- Far worse when tilted from image plane, it typically becomes blocky and jaggy





No unjustified 3D example: Time-series data

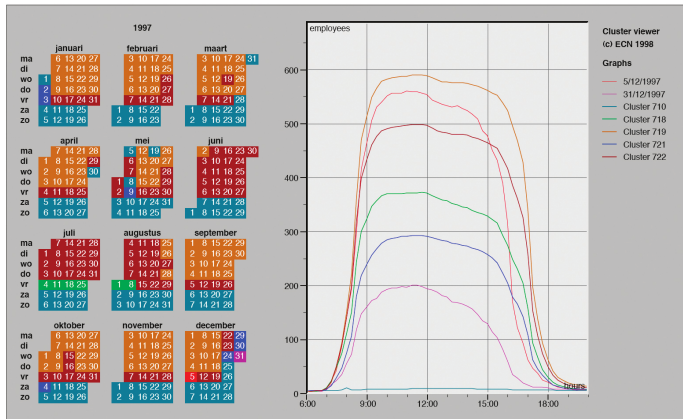
- Extruded curves: detailed comparisons impossible



Transform for new data abstraction



- Derived data: cluster hierarchy
- Juxtapose multiple views: calendar, superimposed 2D curves

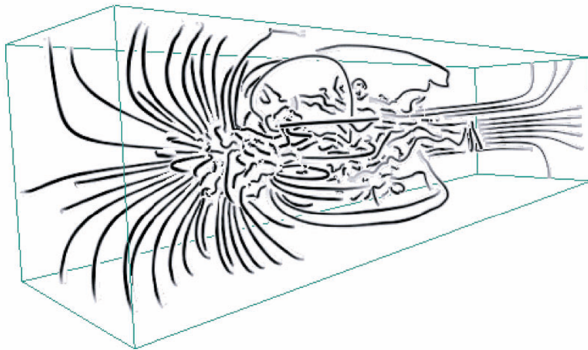




Shape Perception

The great benefit of using 3D comes when the viewer's task fundamentally requires understanding the three-dimensional geometric structure of objects or scenes

- interactive navigation supports synthesis across many viewpoints





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- Experimental evidence suggests that 3D interfaces are better for shape understanding, whereas 2D are best for relative position tasks
- 3D legitimate for true 3D spatial data
- 3D needs very careful justification for **abstract data**
 - enthusiasm in 1990s, but now skepticism
 - be especially careful with 3D for point clouds or networks



No unjustified 2D



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- Consider whether network data requires 2D spatial layout
 - especially if reading text is central to task!
 - arranging as network means lower information density and harder label lookup compared to text lists
- Benefits outweigh costs when topological structure/context important for task
 - be especially careful for search results, document collections, ontologies

➔ Network Data

➔ Topology



➔ Paths





Resolution over immersion



Resolution beat immersion

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- **Immersion**, the feeling of presence in virtual reality, typically not helpful for abstract data
 - do not need sense of presence or stereoscopic 3D
- **Resolution** much more important
 - pixels are the scarcest resource
 - desktop also better for workflow integration
- **Virtual reality** for abstract data very difficult to justify

Overview first, zoom and filter, details on demand





Overview first, zoom and filter, details on demand

- Influential mantra from Shneiderman
 - The need for overview vs. the need to see details
- Overview = summary



Query

→ Identify



→ Compare



→ Summarize



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Responsiveness is required



Responsiveness is required

- The **latency** of interaction, namely, how much time it takes for the system to respond to the user action, matters immensely for **interaction design**

Table 1: Human response to interaction latency changes dramatically at these time thresholds

Time Constant	Value (in seconds)
perceptual processing	0.1
immediate response	1
brief tasks	10

- Importance of visual feedback

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Interactivity Costs

- Interactivity has both power and cost.
- The benefit of interaction is that people can explore a larger information space than can be understood in a single static image.
- However, a cost to interaction is that it requires human time and attention

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Function first, form next



Function first, form next

- Start with focus on functionality
 - straightforward to improve aesthetics later on, as refinement
 - if no expertise in-house, find good graphic designer to work with
- Dangerous to start with aesthetics
 - usually impossible to add function retroactively

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Form: Basic graphic design principles

- **Proximity**

- do group related items together
- avoid equal whitespace between unrelated

- **Alignment**

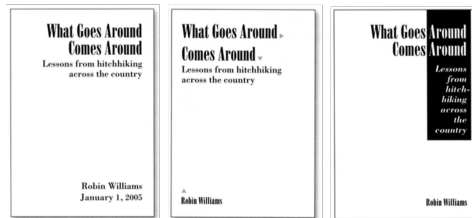
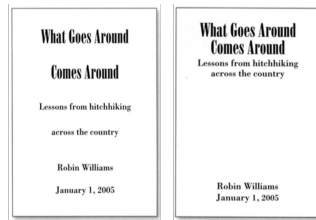
- do find/make strong line, stick to it
- avoid automatic centering

- **Repetition**

- do unify by pushing existing consistencies
- contrast

- If not identical, then very different

- avoid similar



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Memory and Attention

- People have two different categories of memory: long-term memory versus short-term memory also known as **working memory**.
- Human working memory is a very limited resource. When these limits are reached, people experience **cognitive load** and will fail to absorb all of the information that is presented.

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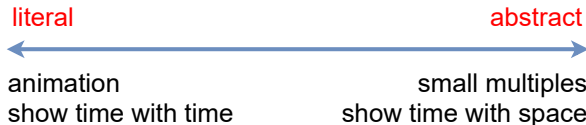
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- Principle: external cognition vs. internal memory
 - easy to compare by moving eyes between side-by-side views
 - harder to compare visible item to memory of what you saw
- Implications for animation
 - great for choreographed storytelling, navigation, complex shape deformations
 - great for transitions between two states: blink comparator
 - poor for many states with changes everywhere (common case when data-driven) → consider small multiples instead

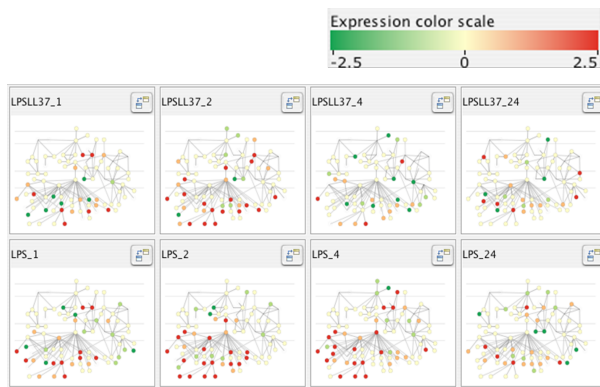


Example: Cerebral



Small multiples: one graph instance per experimental condition

- same spatial layout
- color differently, by condition





Why not animation?

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- Disparate frames and regions: comparison difficult
 - vs contiguous frames
 - vs small region
 - vs coherent motion of group
- Safe special case
- animated transitions



Change blindness

- If attention is directed elsewhere, even drastic changes not noticeable
 - door experiment
- Change blindness demos
 - mask in between images

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